Local Projections in Macroeconomics

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Alessandri, Jordà & Venditti (2025)

- Financial markets are a key component of the monetary policy transmission mechanism.
- Recent evidence suggests monetary shocks have state-dependent effects:
  - effects vary over time with the nature of the intervention;
  - effects depend on the business-cycle state.
- This paper connects these facts: **asymmetric financial-market responses** help explain why **monetary expansions and contractions** have different macro implications.

Empirical I

LP specification:

$$y_{t+h-1} - y_{t-1} = \alpha_h \, \mathrm{MP}_t + \alpha_h^+ \, \mathrm{MP}_t \times \mathrm{I}(\mathrm{MP}_t > 0) + \beta_h(L)' X_{t-1} + \eta_{t,h}$$

Gonçalaves et al. (2021) estimator:

$$\mathrm{IRF}_{h,\delta}^{\mathrm{LP}} = \alpha_h \delta + \alpha_h^+ \mathbf{E}\left[(\mathrm{MP}_t + \delta)^+ - \mathrm{MP}_t^+\right]$$

Notation: $\mathrm{MP}_t^+ = \mathrm{MP}_t \times \mathrm{I}(\mathrm{MP}_t > 0)$

Intuition I

Abstracting from controls and long-difference specification, the regression becomes:

$$y_{t+h} = \alpha_h \, \mathrm{MP}_t + \alpha_h^+ \, \mathrm{MP}_t^+ + \eta_{t,h}$$

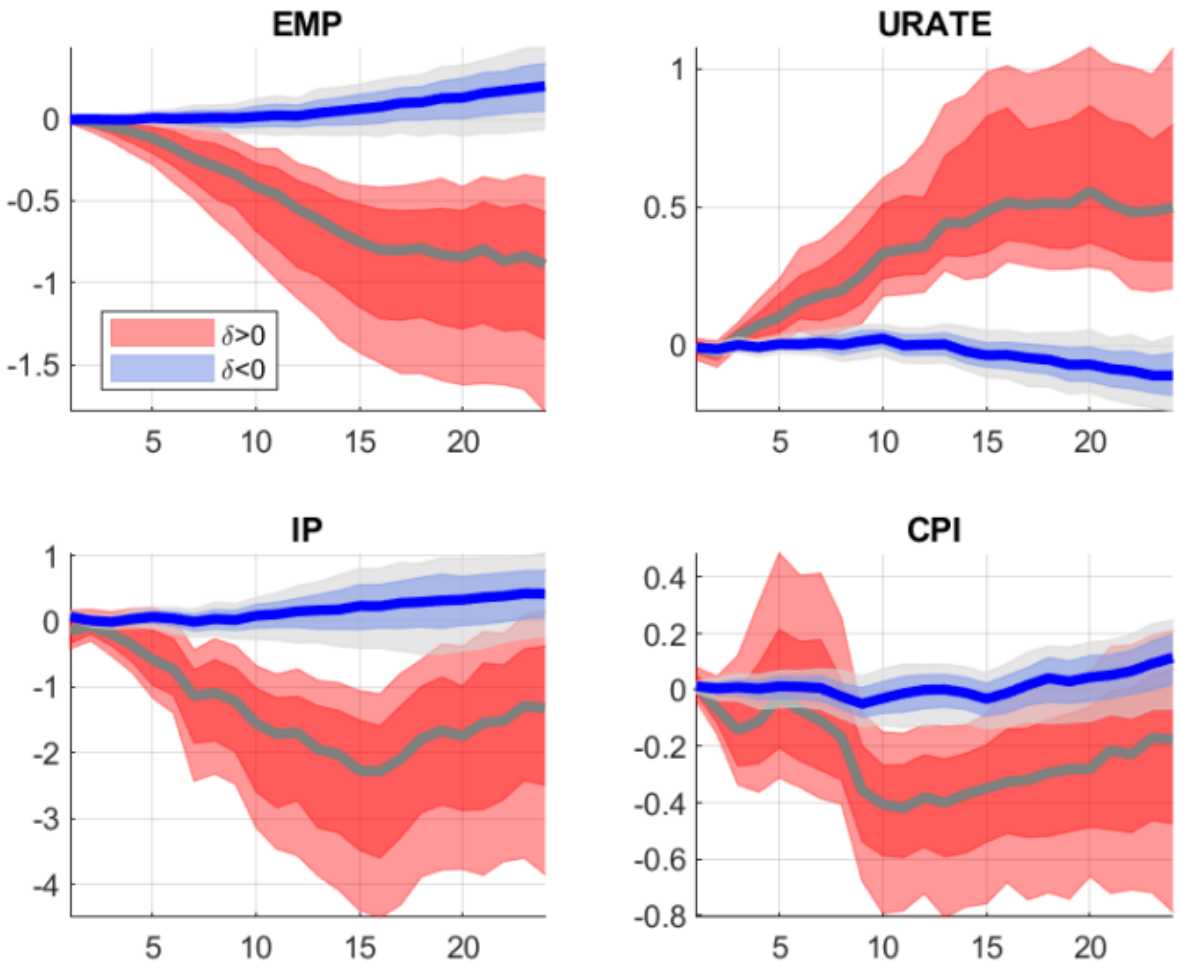
Then:

$$\left.\frac{\partial y_{t+h}}{\partial \mathrm{MP}_t}\right|_{\mathrm{MP}_t > 0} = \alpha_h + \alpha_h^+$$

and:

$$\left.\frac{\partial y_{t+h}}{\partial \mathrm{MP}_t}\right|_{\mathrm{MP}_t < 0} = \alpha_h$$

Results I


EMP
0
-0.5
-1
-1.5
δ>0
δ<0
5
10
15
20
URATE
1
0.5
0
5
10
15
20
IP
1
0
-1
-2
-3
-4
5
10
15
20
CPI
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
5
10
15
20

Empirical II

Obtain the fitted values $\widehat{\mathrm{EBP}}_t$ from:

$$\mathrm{EBP}_t = \theta\,\mathrm{UNC}_t + \Gamma' X_{t-1} + \omega_t$$

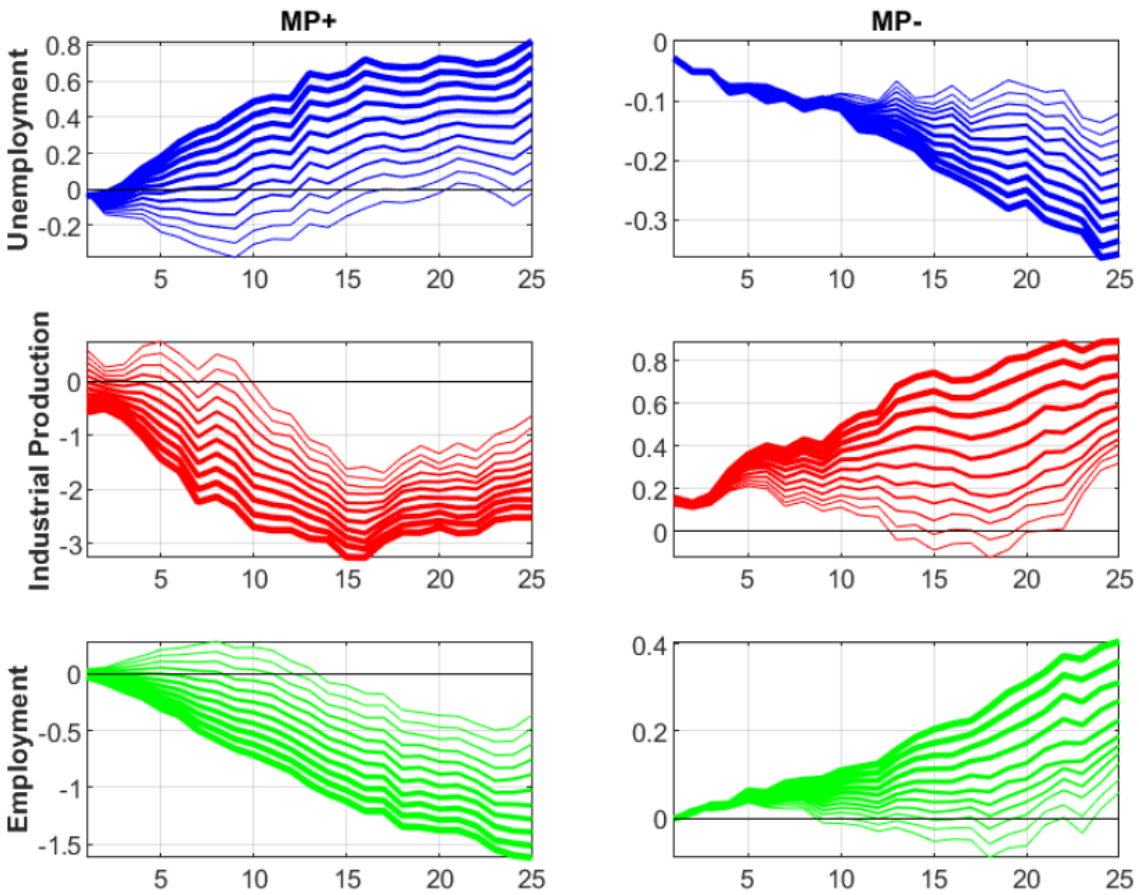
Use $\widehat{\mathrm{EBP}}_t$ in the LP specification:

$$\begin{aligned} y_{t+h-1} - y_{t-1} = \alpha_h\,\mathrm{MP}_t + \alpha_h^+\,\mathrm{MP}_t^+ + \beta_h\,\mathrm{MP}_t\widehat{\mathrm{EBP}}_t \\ + \beta_h^+\,\mathrm{MP}_t^+\widehat{\mathrm{EBP}}_t + \beta_h(L)' X_{t-1} + \eta_{t,h} \end{aligned}$$

Gonçalaves et al. (2021) estimator:

$$\begin{aligned} \mathrm{IRF}_{h,\delta}^{\mathrm{LP}} = \alpha_h\delta + \alpha_h^+\mathbf{E}\left[(\mathrm{MP}_t+\delta)^+ - \mathrm{MP}_t^+\right] \\ + \widehat{\mathrm{EBP}}_t\left[\beta_h\delta + \beta_h^+\mathbf{E}\left[(\mathrm{MP}_t+\delta)^+ - \mathrm{MP}_t^+\right]\right] \end{aligned}$$

Results II


MP+
MP-
Unemployment
Industrial Production
Employment
0.8
0.6
0.4
0.2
0
-0.2
-0.1
-0.3
-1
-2
-3
-0.5
-1.5
5
10
15
20
25

Appendix – Variance Derivation

$$[1, 1] \begin{bmatrix} \sigma_x^2 & \sigma_{xy} \\ \sigma_{xy} & \sigma_y^2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} \sigma_x^2 + \sigma_{xy} & \sigma_{xy} + \sigma_y^2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= (\sigma_x^2 + \sigma_{xy}) + (\sigma_{xy} + \sigma_y^2)$$

$$= \sigma_x^2 + \sigma_y^2 + 2\sigma_{xy}$$